HESOD: HIGH-RESOLUTION EFFICIENT SMALL OBJECT DETECTION VIA SEMANTIC-SPECTRAL DUAL SELECTION

Anonymous Authors

Affiliation(s) omitted for blind review

ABSTRACT

Detecting tiny objects in high-resolution aerial and maritime imagery is severely challenged by massive background redundancy. Spatial selective computation mitigates arithmetic overhead by routing candidate foreground regions; however, existing selectors rely exclusively on deep semantic activations, which frequently collapse for micro targets, leading to irrecoverable early-stage pruning. We propose **HESOD** (**H**igh-Resolution **E**fficient **S**mall **O**bject **D**etection), a selective framework featuring semantic-spectral dual-evidence routing and coverage-constrained spatial optimization. HESOD couples high-level semantic objectness with channel-pooled spectral saliency at $\mathcal{O}(1)$ channel complexity to robustly rescue weak targets. To guarantee recall safety, an object-level soft-coverage loss directly optimizes multi-cell joint coverage probability across all ground-truth instances. Combined with an inverted residual partial-convolution head and scale-aware Wasserstein regression, HESOD improves $\mathrm{mAP@.5}$ by +2.4 pp on SeaPerson while cutting parameters by 27.6% at 80.7 FPS.

Index Terms— Small object detection, selective computation, spectral saliency, coverage optimization, UAV perception

1. INTRODUCTION

High-resolution aerial and maritime vision is essential for remote surveillance, environmental monitoring, and search-and-rescue [1, 2, 4]. In these settings, micro targets typically occupy fewer than 16×16 pixels amidst vast background areas ($>$ 70–80%), making dense uniform convolution computationally prohibitive on edge platforms [3, 5].

Spatial selective computation mitigates arithmetic redundancy by routing processing strictly to candidate foreground regions [3, 6, 7]. However, existing selectors exhibit two critical vulnerabilities: (i) *Evidence fragility*: relying solely on deep semantic activations leads to severe feature collapse for micro targets after stem downsampling, causing irrecoverable false-negative pruning under asymmetric selection cost [3, 8]; and (ii) *Coverage deficit*: standard per-pixel binary cross-entropy optimizes grid cells independently, failing to guarantee joint multi-cell coverage over individual instances.

To address these limitations, we present **HESOD** (**H**igh-Resolution **E**fficient **S**mall **O**bject **D**etection), an efficient selective detection framework. Our key contributions are:

1. **Dual-Evidence Spatial Selector**: We couple high-level semantic objectness with channel-pooled spectral saliency at $\mathcal{O}(1)$ channel complexity, reliably preserving degraded micro targets.
2. **Object-Level Soft-Coverage Loss** ($\mathcal{L}_{\text{cover}}$): We formulate a differentiable joint-coverage objective to prevent premature false-negative pruning under asymmetric selection cost.
3. **Edge-Efficient Decoupled Architecture**: By integrating an inverted residual ISPP head with scale-aware SABL regression, HESOD establishes an optimal accuracy-efficiency frontier (-27.6% params at 80.7 FPS).

2. RELATED WORK

2.1. Small Object Detection in Aerial Imagery

Detecting micro targets in high-resolution aerial and maritime vision is challenged by extreme scale disparity, low contrast, and massive background areas [1, 2, 4, 10]. Classical solutions employ density-guided patch cropping (e.g., ClusDet [11], DMNet [12], UFPMP-Det [13]) or multi-stage coarse-to-fine slicing (e.g., ResBiDet [14]). While focusing compute, these approaches incur redundant feature extraction across overlapping tiles and complex boundary stitching. Other works design specialized label assignment and regression metrics, such as Normalized Gaussian Wasserstein Distance (NWD) [15], RFLA [16], and scale-aware balancing (SSABNet) [9]. However, dense detectors uniformly evaluate all spatial locations, leaving background arithmetic redundancy unresolved.

2.2. Selective and Dynamic Computation

Dynamic computation reduces arithmetic overhead by evaluating only prospective foreground regions. AutoFocus [5] predicts focus pixels for multi-scale inference; QueryDet [6] accelerates feature pyramids through cascaded sparse queries;

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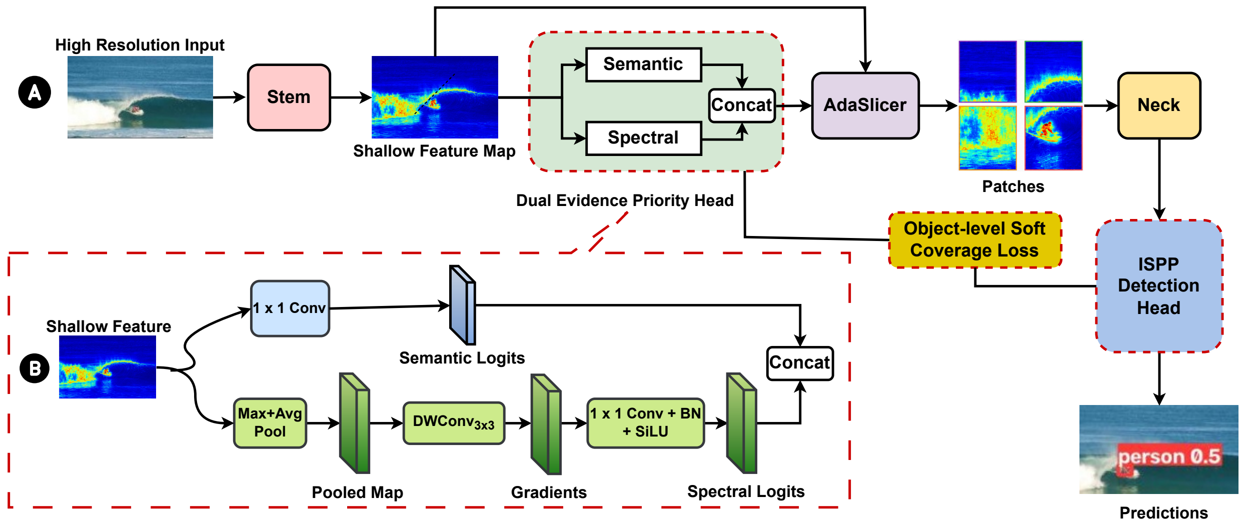

Fig. 1. Overview of HESOD: (a) End-to-end selective pipeline; (b) Dual-evidence priority head. Red dashed boxes indicate our proposed contributions.

and CEASC [7] designs adaptive activation masks. ESOD [3] introduced an early-stage spatial selector after the stem to route foreground patches (AdaSlicer) to downstream stages. Nonetheless, existing selectors depend strictly on single semantic activations and independent per-cell BCE loss, leaving them vulnerable to early-stage pruning. In contrast, HESOD operates in a unified feature-space selective pipeline, combining dual-evidence routing with joint soft-coverage optimization.

2.3. Spectral Saliency and Frequency Representations

Frequency-domain cues provide rich spatial structural priors. SET [8] demonstrated that micro objects preserve distinct high-frequency gradients even when semantic responses fade, though it applies dense frequency operations across feature pyramids. HESOD instead repurposes spectral saliency as an ultra-lightweight, channel-pooled routing signal to ensure recall-safe selection at negligible overhead.

3. PROPOSED METHOD

The overall architecture of HESOD is illustrated in Fig. 1(a). Given an input image $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$, a shallow stem extracts dense feature map $\mathbf{F} \in \mathbb{R}^{H_s\times W_s\times C}$. The *Dual-Evidence Spatial Selector* then proceeds in two steps: (i) a priority head fuses semantic objectness with channel-pooled spectral saliency (Fig. 1(b)) to generate a continuous priority map $\mathbf{S} \in [0,1]^{H_s\times W_s}$; and (ii) an AdaSlicer module dynamically crops high-priority candidate patches $\mathcal{P}$ directly from $\mathbf{F}$. Downstream backbone, neck, and lightweight ISPP decoupled heads evaluate only these retained patches, supervised end-to-end by an object-level soft-coverage loss and scale-aware regression (§3.2, §3.3).

3.1. Dual-Evidence Priority Estimation

To rescue targets with collapsed semantic responses, HESOD couples semantic confidence with structural gradient saliency (Fig. 1(b)).

Let $\mathbf{f}_i \in \mathbb{R}^C$ denote the feature vector at spatial location i in $\mathbf{F}$. The semantic branch predicts class logits $\mathbf{z}_i^{\text{sem}} = \text{Conv}_{1\times1}(\mathbf{f}_i) \in \mathbb{R}^{N_c}$. In parallel, the spectral branch extracts high-frequency structural gradients by compressing channels via max- and mean-pooling:

$$\mathbf{F}_{\text{pooled}} = \left[\max_c \mathbf{F}_{:,:,c} \,\|\, \frac{1}{C}\sum_{c=1}^{C}\mathbf{F}_{:,:,c}\right] \in \mathbb{R}^{H_s\times W_s\times 2}, \quad (1)$$

where $[\cdot \,\|\, \cdot]$ denotes concatenation. Compressing $\mathbf{F}$ to 2 channels guarantees that subsequent spatial filtering complexity remains $\mathcal{O}(1)$ regardless of backbone width C.

Next, a learnable depthwise 3×3 convolution bank, initialized with discrete Laplacian and directional Sobel filters, is applied over $\mathbf{F}_{\text{pooled}}$ to capture isotropic edge transitions and directional gradients. A 1×1 projection layer maps the 6 filtered channels to dimension N_c, yielding spectral logits $\mathbf{z}_i^{\text{spec}}$. Finally, semantic and spectral logits are concatenated and fused via a 1×1 convolution to predict class-agnostic routing priority scores $s_i \in [0,1]$:

$$s_i = \sigma(\text{Conv}_{1\times1}([\mathbf{z}_i^{\text{sem}} \,\|\, \mathbf{z}_i^{\text{spec}}]))\,. \quad (2)$$

Candidate regions with $s_i > \tau$ are clustered into bounding patches and routed downstream.

3.2. Object-Level Soft-Coverage Loss

Standard pixel-wise binary cross-entropy (BCE) evaluates grid cells independently, failing to guarantee instance coverage: a selector may over-allocate cells to large objects while

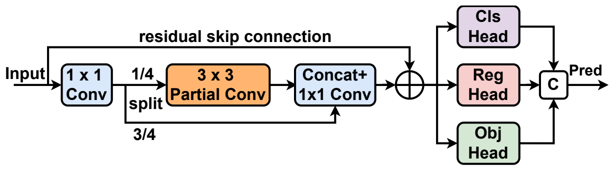

Fig. 2. Inverted Residual Decoupled Head (ISPP).

dropping tiny objects whose receptive fields contain only weak responses. To prevent premature false negatives, we formulate a differentiable object-level soft-coverage loss.

Let $\mathcal{N}(j)$ denote the set of selector cells overlapping ground-truth box $j \in \{1, \ldots, N_{\text{gt}}\}$. Assuming independent cell activations, the probability that instance j is covered by at least one selected cell is:

$$p_j^{\text{cover}} = 1 - \prod_{i \in \mathcal{N}(j)} (1 - s_i). \tag{3}$$

The soft-coverage loss directly optimizes joint instance coverage:

$$\mathcal{L}_{\text{cover}} = -\frac{1}{N_{\text{gt}}} \sum_{j=1}^{N_{\text{gt}}} w_j \log\left(p_j^{\text{cover}} + \epsilon\right), \tag{4}$$

where $w_j = \text{clip}(4/a_j, 1, 5)$ is a scale-adaptive weight based on bounding box area a_j in grid cells, penalizing misses on micro targets ($a_j < 4$). The total training objective is $\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda_{\text{sel}}(\mathcal{L}_{\text{BCE}} + \lambda_{\text{cov}}\mathcal{L}_{\text{cover}})$, with $\lambda_{\text{sel}} = 0.2$ and $\lambda_{\text{cov}} = 0.5$.

3.3. Efficient Decoupled Head and Scale-Aware Loss

To reduce computational overhead while boosting localization precision, we incorporate an inverted residual partial-convolution (ISPP) decoupled head [17] and a Scale-Aware Balancing Loss (SABL) [9].

As shown in Fig. 2, the ISPP head expands neck feature channels to $2C_1$, partitioning them into an active branch (0.25 ratio) processed by a 3×3 Partial Convolution (PConv) and an identity passthrough branch (0.75 ratio), where Ⓒ denotes channel concatenation. After linear projection and residual addition, decoupled 1×1 heads predict classification, box regression, and objectness independently, drastically lowering compute while eliminating task interference.

Furthermore, standard IoU-based regression suffers from vanishing gradients on micro objects due to extreme sensitivity to tiny spatial misalignments. To ensure robust localization, SABL dynamically balances Normalized Gaussian Wasserstein Distance (D_W) and Complete IoU (CIoU) via scale-dependent gating $\mu(s) = \exp(-(s/32)^6)$:

$$\mathcal{L}_{\text{box}} = 1 - \text{IoU} + \mu(s)\left(1 - e^{-D_W/12}\right) + (1 - \mu(s))\ell_{\text{ctr}} + \alpha v, \tag{5}$$

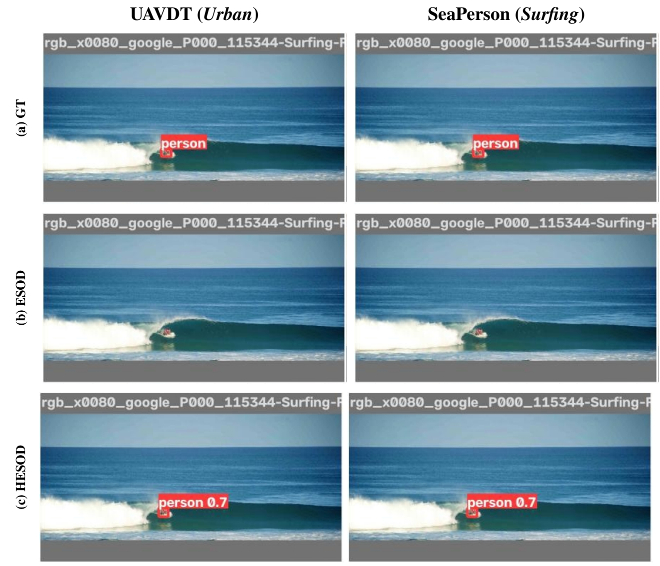

Fig. 3. Qualitative comparison on UAVDT and SeaPerson.

where $s = \sqrt{w \cdot h}$ is the bounding box geometric scale, ℓ_{ctr} penalizes normalized center distance, and v measures aspect ratio consistency. For micro targets ($s \ll 32$ px), $\mu(s) \to 1$, activating the non-vanishing Wasserstein distribution distance to provide stable continuous gradients even under zero spatial IoU overlap; for larger objects ($s \gg 32$ px), $\mu(s) \to 0$, smoothly reverting to standard CIoU regression.

4. EXPERIMENTS

4.1. Experiment Settings

We evaluate HESOD on two demanding high-resolution small-object benchmarks: **UAVDT** [2] (1280×1280 px, urban vehicle surveillance across 3 classes) and **SeaPerson** [4, 10] (2048×2048 px, maritime search-and-rescue with $> 85\%$ micro targets < 32 px). Both benchmarks evaluate full-resolution images without artificial pre-cropping.

All models are implemented in PyTorch using a YOLOv5m base detector and trained with SGD for 50 epochs following standard selective protocols [3]. Detection accuracy is assessed via COCO-style mAP@.5 and scale-stratified recall across pixel area bins: *Very Tiny* ($< 16^2$), *Tiny* (16^2–32^2), and *Small* (32^2–96^2). Computational efficiency is measured via parameter count, GFLOPs, FPS (batch size 1), and Bounding Patch Recall (BPR).

4.2. Main Benchmark Comparisons

Comparison with Dense Baselines. As shown in Table 1, classical dense detectors incur prohibitive arithmetic overhead on multi-megapixel vision grids due to uniform convolution. Compared to Faster R-CNN and RetinaNet on SeaPerson (2048×2048), HESOD delivers substantial accuracy gains of **+22.3 pp** and **+30.1 pp** mAP@.5, while slashing

Table 1. Benchmark results on UAVDT and SeaPerson.

Method	mAP @.5	mAP @.5:.95	Params (M)	GFLOPs	FPS
UAVDT Benchmark *(1280 × 1280)*					
Faster R-CNN [18]	*[TBD]*	*[TBD]*	41.5	*[TBD]*	*[TBD]*
RetinaNet [19]	*[TBD]*	*[TBD]*	34.0	*[TBD]*	*[TBD]*
ESOD [3]	**0.385**	**0.214**	<u>35.9</u>	**68.2**	**117.8**
HESOD (Ours)	<u>0.378</u>	<u>0.202</u>	**25.9**	<u>81.6</u>	<u>95.2</u>
Delta (Δ)	-0.7 pp	-1.2 pp	**-27.6%**	+13.4	-22.6
SeaPerson Benchmark *(2048 × 2048)*					
Faster R-CNN [18]	0.551	0.246	43.3	1546.8	23.1
RetinaNet [19]	0.473	0.201	36.4	942.7	28.0
ESOD [3]	<u>0.750</u>	<u>0.320</u>	<u>35.8</u>	**202.4**	**85.7**
HESOD (Ours)	**0.774**	**0.326**	**25.9**	<u>209.0</u>	<u>80.7</u>
Delta (Δ)	**+2.4 pp**	**+0.6 pp**	**-27.6%**	+6.6	-5.0

Bold: best; <u>underline</u>: second-best.

Table 2. Module ablation on UAVDT and SeaPerson.

Arm	Selector	Box Loss	Head	mAP @.5	GFLOPs	Very Tiny	Tiny	Small	Total
UAVDT Benchmark *(1280 × 1280)*									
(1)	Sem	CIoU	Cpl	0.385	**68.2**	79.43%	84.21%	94.54%	85.17%
(2)	Sem*	CIoU	Cpl	0.384	<u>75.0</u>	79.13%	83.91%	94.00%	84.82%
(3)	Spec*	CIoU	Cpl	**0.396**	98.1	83.78%	87.83%	97.62%	88.83%
(4)	Spec-P*	CIoU	Cpl	<u>0.394</u>	99.3	**86.88%**	**92.23%**	97.74%	**91.93%**
(5)	Dual*	CIoU	Cpl	0.371	83.9	79.09%	85.85%	95.85%	86.35%
(6)	Dual*	SABL	Cpl	0.360	97.1	<u>83.53%</u>	<u>89.43%</u>	97.93%	<u>89.72%</u>
(7)	Dual*	CIoU	ISPP	0.371	82.6	83.06%	88.19%	<u>98.07%</u>	88.94%
(8)	Dual*	SABL	ISPP	0.378	81.6	82.54%	88.34%	**98.09%**	88.94%
SeaPerson Benchmark *(2048 × 2048)*									
(1)	Sem	CIoU	Cpl	0.750	**202.4**	74.03%	86.78%	94.74%	84.42%
(2)	Sem*	CIoU	Cpl	0.769	266.8	75.49%	91.57%	95.71%	87.75%
(3)	Spec*	CIoU	Cpl	0.770	267.4	75.23%	**91.69%**	**95.89%**	87.77%
(4)	Spec-P*	CIoU	Cpl	0.767	263.4	76.13%	91.36%	95.50%	87.77%
(5)	Dual*	CIoU	Cpl	<u>0.772</u>	281.2	76.00%	91.50%	95.68%	87.84%
(6)	Dual*	SABL	Cpl	0.771	263.7	<u>76.67%</u>	91.49%	94.77%	<u>87.89%</u>
(7)	Dual*	CIoU	ISPP	0.771	217.6	75.86%	91.25%	<u>95.85%</u>	87.68%
(8)	Dual*	SABL	ISPP	**0.774**	<u>209.0</u>	**77.14%**	<u>91.59%</u>	95.01%	**88.11%**

*Trained with $\mathcal{L}_{\text{cover}}$ (vs. standard BCE). Arm (8): full HESOD. Selectors: Sem (Semantic), Spec (Spectral), Spec-P (Spectral-Pooled), Dual (Dual-Concat). Head: Cpl (Coupled), ISPP (Ours). Scale bins: Very Tiny ($< 16^2$), Tiny (16^2–32^2), Small (32^2–96^2 px^2). **Bold**: best; <u>underline</u>: second-best.

GFLOPs by **7.4**× and **4.5**× at **80.7 FPS**, confirming that selective computation is essential for edge deployment.

Comparison with Selective Baseline (ESOD). Against ESOD, HESOD achieves clear accuracy, recall, and parameter advantages across both benchmarks. On SeaPerson, HESOD advances $\mathrm{mAP@.5}$ by **+2.4 pp** (0.774 vs. 0.750) and $\mathrm{mAP@.5:.95}$ by **+0.6 pp**, while cutting parameters by **27.6%** (25.9M vs. 35.8M) at 80.7 FPS. On UAVDT (1280×1280), HESOD enhances overall instance recovery by **+3.77 pp** (88.94% vs. 85.17%, recovering +14,109 targets) with consistent recall gains across all vehicle classes (*car*: +3.70 pp, *truck*: +2.85 pp, *bus*: +7.96 pp), establishing a superior accuracy-efficiency Pareto frontier.

4.3. Ablation Studies and Analysis

Dual Evidence and Soft-Coverage Optimization. Table 2 isolates individual module contributions. On SeaPerson, replacing standard BCE with object-level soft-coverage loss $\mathcal{L}_{\text{cover}}$ (Arm $1 \to 2$) yields an immediate **+1.9 pp** $\mathrm{mAP@.5}$ boost and raises Bounding Patch Recall (BPR) to **0.991**, verifying that joint coverage prevents premature selector pruning. Furthermore, channel-pooled spectral routing (Arm 4) preserves unpooled accuracy (Arm 3) at $\mathcal{O}(1)$ spatial complexity, while dual-evidence fusion (Arm 5) attains **0.772** $\mathrm{mAP@.5}$ and rescues +631 micro targets ($< 16^2$ px).

Edge Decoupled Head and Localization Loss. Adopting the inverted residual ISPP head (Arm 7) cuts compute by **22.6%** (217.6 vs. 281.2 GFLOPs) and trims parameters by **27.6%**, achieving **0.328** $\mathrm{mAP@.5:.95}$. Adding scale-aware Wasserstein regression (SABL, Arm 8) further improves sub-16px localization, pushing *Very Tiny* recall to a peak **77.14%** (+2,560 targets over baseline), overall recall to **88.11%**, and runtime compute to **209.0 GFLOPs** at **80.7 FPS**. Full HESOD thus attains the optimal Pareto trade-off.

Error Bottleneck Diagnosis. Stratified breakdown confirms that target recovery is strongest on *Tiny* (+4.81 pp) and *Very Tiny* (+3.11 pp) strata. Error diagnosis reveals that dual routing and coverage optimization reduce selector-dropped false negatives from 25.6% in baseline ESOD to 19.2%, shifting the remaining bottleneck to downstream refinement.

Qualitative Comparison. As shown in Fig. 3 on UAVDT and SeaPerson, baseline ESOD misses weak-contrast micro targets due to semantic degradation, whereas HESOD's dual-evidence routing and scale-aware regression accurately localize all tiny instances without false alarms.

5. CONCLUSION

In this paper, we presented HESOD, an efficient dual-evidence selective framework for small object detection in high-resolution aerial and maritime imagery. By coupling semantic objectness with channel-pooled spectral saliency at $\mathcal{O}(1)$ complexity and optimizing joint instance coverage via an object-level soft-coverage loss ($\mathcal{L}_{\text{cover}}$), HESOD effectively prevents premature false-negative pruning on micro targets. Integrated with an inverted residual partial-convolution head (ISPP) and scale-aware Wasserstein regression (SABL), HESOD achieves +2.4 pp $\mathrm{mAP@.5}$ on SeaPerson and +3.77 pp recall on UAVDT over baseline ESOD, while cutting parameters by 27.6% at 80.7 FPS.

Future work will investigate adaptive spatial threshold scheduling and extend dual-evidence selective routing to multimodal aerial perception platforms.

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